

Dataset v1.3 — In Progress

n=32 verified · Target n=50 before next DOI release · Framework v1.2

DOI v1.2: 10.5281/zenodo.20270575 · License: CC-BY-NC 4.0 · zorthex.com · github.com/zorthex2026/zorthex-diffusion-lag

// WHAT IS ZORTHEX

Framework Overview

Zorthex is an independent research practice measuring the temporal gap between operational emergence and institutional legibility in financial and technological infrastructure transitions.

$L = t_{\text{peak}} - t_{\text{start}}$ (months) — the observable gap between when something becomes operationally real and when public attention structurally consolidates around it.

Zorthex is descriptive only and does not constitute investment advice.

// CORE DEFINITIONS

Locked Across All Versions

- t_{start} — first documented operational emergence, justified with primary source (Level A–D)
- t_{peak} — first month of 12 consecutive months $\geq 25/100$ on Google Trends Worldwide
- $L = t_{\text{peak}} - t_{\text{start}}$ in months
- STRUCTURAL — $\geq 25/100$ for 12+ consecutive months (permanent once acquired)
- OBSERVATION — above threshold, fewer than 12 consecutive months
- BUBBLE — peaked above threshold, dropped below 25, no 12-month window (permanent)

// 4-REGIME DIFFUSION TAXONOMY

Empirical — n=32

Regime	Name	Pattern	Examples
1	Policy-Trigger	Slow ramp → regulatory breakout	PQC (91mo), Stablecoins (128mo)
2	Institutional Mass	Multi-year invisibility → explosive breakout	RWA (103mo), EV Infra
3	Market-Narrative	Speculation cycles, media amplification	Bitcoin (107mo), Crypto
4	Shock/Spoke	Single-event spike → memory-zero collapse	GDPR, NFT partial

Taxonomy is a working hypothesis — to be verified at n=50.

// NON-STATIONARITY FLAG

Mandatory Per Case

State	Description	L Comparable?
STABLE	Organic diffusion, no exogenous shock	Yes — fully
SEMI-STABLE	Partial external influence documented	With caution
NON-STATIONARY	Exogenous shock dominant (e.g. COVID)	No — excluded from averaging

// LAYER A — HARD INFRASTRUCTURE (18 cases, L-comparable)

Layer A Dataset

Phenomenon	L (mo)	Classification	Regime	Stationarity
Stablecoins ★	128	STRUCTURAL	Policy	STABLE
RWA Tokenization ★	103	OBSERVATION	Inst. Mass	STABLE
Post-Quantum Crypto ★	91	STRUCTURAL	Policy	STABLE
Quantitative Easing	136	STRUCTURAL	Policy	STABLE
Bitcoin	107	STRUCTURAL	Market	STABLE
Cryptocurrency	96	STRUCTURAL	Market	STABLE
CBDC	~89	STRUCTURAL	Policy	STABLE
Tokenized Deposits	~78	OBSERVATION	Inst. Mass	STABLE
Quantum Computing	~113	OBSERVATION	Inst. Mass	STABLE
Edge Computing	~120	STRUCTURAL	Inst. Mass	STABLE
SMR	199	OBSERVATION	Inst. Mass	STABLE
Green Hydrogen	~198	STRUCTURAL	Policy	STABLE
Carbon Capture	~204	STRUCTURAL	Policy	STABLE
EV Infrastructure	~154	OBSERVATION	Inst. Mass	STABLE
Liquid Biopsy	~174	STRUCTURAL	Inst. Mass	STABLE
CRISPR	82	STRUCTURAL	Inst. Mass	STABLE
ESG Reporting	~212	STRUCTURAL	Policy	STABLE
Gene Therapy	—	STRUCTURAL	Inst. Mass	STABLE

★ = *Verified Custom Report available at zorthex.com/research.html*

// LAYER B — ATTENTION PHENOMENA (14 cases, regime-classifiable only)

Layer B Dataset

Phenomenon	Classification	Regime	Stationarity
mRNA Cancer Vaccines	OBSERVATION	Policy	SEMI-STABLE
LLM / GenAI	STRUCTURAL	Market	SEMI-STABLE
Deep Learning	STRUCTURAL	Inst. Mass	STABLE
Autonomous Vehicle	STRUCTURAL	Inst. Mass	STABLE
EU AI Act	OBSERVATION	Policy	STABLE
iPhone	STRUCTURAL	Market	STABLE
Cloud Computing	STRUCTURAL	Market	STABLE
Facebook / Social Media	STRUCTURAL	Market	STABLE
Virtual Reality	STRUCTURAL	Market	STABLE
Remote Work	STRUCTURAL	Market	NON-STATIONARY
DeFi Summer	OBSERVATION	Market	SEMI-STABLE
NFT	BUBBLE	Shock	SEMI-STABLE
Metaverse	BUBBLE	Shock	SEMI-STABLE
GDPR	BUBBLE	Shock	NON-STATIONARY

// VERSION CHANGELOG

Release History

Version	DOI	Date	Notes
v1.0	10.5281/zenodo.20049068	Apr 2026	Initial, n=11, 3-month window
v1.1	10.5281/zenodo.20072999	May 2026	12-month window, BUBBLE, n=12
v1.2	10.5281/zenodo.20270575	May 2026	t_start policy, CSV locking, 4-tier
v1.3	pending — target n=50	Jun 2026	4-regime, Layer A/B, Non-Stationarity

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If you follow this pipeline and obtain different results — that is a contribution, not a problem.